Supplementary Material for "Towards Trustworthy IoT Network Security: Interpretable Open-Set Intrusion Detection"

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This supplement provides the complete concept and rule definitions, the CTU-IoT-23 feature derivation, the threshold-gradient derivation, expanded ablations, per-class and per-family results, and additional robustness analyses. The main article presents the methods, principal evidence, and limitations supporting the central conclusions. Table, figure, and section numbers here are prefixed "S"; references to the main article use its numbering.

Feature set. Following the known-class feature-selection control (Section SVII), all CTU-IoT-23 results in the main article and in the finding-bearing supplementary tables (this control, the α-gate sweep and head-agreement, the per-family open-set breakdown, the concept-accuracy and leakage probe, the per-class table and the temporal split) use the leak-free known-class-only feature selection. Two supporting analyses below, the λ-sweep and the CTU half of the (λ, γ) grid, report the original 37-feature runs as an ablation; their conclusions are qualitative and unaffected by the feature list, and the paired original-vs-known-only comparison is in Table S6. Every split is drawn deterministically by the released scripts, and the released code includes a script that prints every table value from the released results.

TABLE S1: Branch agreement across the λ_α sweep (NeSy-NIDS, mean over five seeds): learned α and the fraction of flows on which the deployed prediction equals the neural-only (dep=neural) or rule-only (dep=rule) prediction.

λ_α	0	0.05	0.1	0.2	0.5	1.0
CTU α	0.38	0.58	0.69	0.81	0.86	0.89
CTU dep=neural	1.00	1.00	1.00	1.00	1.00	1.00
CTU dep=rule	0.32	0.36	0.38	0.48	0.51	0.48
CIC α	0.32	0.81	0.86	0.90	0.93	0.95
CIC dep=neural	0.99	0.98	0.97	0.98	0.98	0.97
CIC dep=rule	0.45	0.52	0.53	0.53	0.52	0.55

TABLE S2: α-gate sweep (NeSy-NIDS, mean over five seeds): learned α and OOD AUROC per λ_α.

λ_α	0	0.05	0.1	0.2	0.5	1.0
CTU α	0.38	0.58	0.69	0.81	0.86	0.89
CTU AUROC	0.904	0.900	0.909	0.900	0.899	0.893
CIC α	0.32	0.81	0.86	0.90	0.93	0.95
CIC AUROC	0.657	0.679	0.691	0.675	0.675	0.684

SI. The α-Gate and Interpretability Trade-Off

We sweep λ_α over six values, five seeds each (Table S2). The regulariser raises the learned gate *monotonically*. On CTU-IoT-23, α rises from 0.38 (λ_α=0) to 0.89 (λ_α=1.0); on CIC-IoT-2023, from 0.32 to 0.95. F1 does not move: the known-class *validation* F1 (the model-selection split reported in this sweep) stays near 0.933 on CTU and 0.818–0.819 on CIC; the corresponding held-out *test* F1 is 0.933 on CTU (Table III). OOD AUROC is flat on CTU (within 0.016 across the sweep) and rises on CIC, from 0.657 at λ_α=0 to 0.675–0.691 for every $\lambda_\alpha \geq 0.05$, a gain of 0.02–0.03 (about one seed std) that accompanies the increase in α; as the next paragraph shows, it is not evidence that the decision moved onto the rules.

Does raising α actually move the decision onto the rules? No. A larger gate weight need not mean more rule reliance, because the two heads can emit logits of different magnitudes and the learned weights can compensate. Testing reliance directly by branch agreement (Table S1), the deployed prediction stays with the *neural-only* head at every λ_α, and its agreement with the *rule-only* head rises only mildly as α increases (CTU 32% to 51%), staying well below neural agreement. So λ_α tunes the gate scalar but not decision-level reliance on the rules; we describe α as the learned rule-head weight in the logit blend, not as a measure of how interpretable the decision is. (This is also a global scalar gate, so it cannot route individual hard flows to the neural branch.) Each point was trained from scratch.

SII. Concept Leakage and the γ Ablation

Table S3 reports mean concept prediction accuracy for each CBM variant over five seeds. This quantity is the *concept accuracy* of the main text: agreement between the bottleneck's predicted concept and the declared threshold label, i.e. how faithfully the model reproduces the declared predicate, not whether the predicate is semantically "correct". JointCBM without concept regularisation (γ=0) reaches 87.4% on CTU and 96.3% on CIC. On CTU, accuracy rises to 94.2% at γ=1.0 (not monotonically: 89.1% at γ=0.1, 88.9% at 0.5); at γ=0 the weakest concept is `is_persistent` at 81% (it rises to 91% by γ=1.0). On CIC it rises from 96.3% to 97.2%, with every concept above 90%. SequentialCBM reaches near-perfect concept accuracy (99.9% / 99.5%) because its concept head is trained on the concept labels alone and frozen before the classifier is trained. HybridCBM is similar (99.6% / 99.2%): its skip connection lets the classifier bypass the bottleneck, so

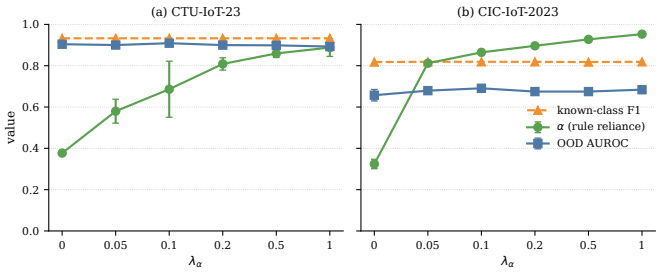


Fig. S1: α-gate sweep (NeSy-NIDS, mean over five seeds; error bars ± 1 std). Raising λ_α drives the learned gate α (the rule *weight* in the logit blend, green) monotonically upward while known-class F1 (orange, dashed) stays flat and OOD AUROC (blue) is flat on CTU-IoT-23 but rises modestly on CIC-IoT-2023.

TABLE S3: Mean concept prediction accuracy (%) on the known-class test split per CBM variant (mean over five seeds; λ sensitivity in Table S4).

Model	**CTU-IoT-23**	**CIC-IoT-2023**
JointCBM (γ=0)	87.4	96.3
JointCBM (γ=0.1)	89.1	96.5
JointCBM (γ=0.5)	88.9	96.8
JointCBM (γ=1.0)	94.2	97.2
SequentialCBM	99.9	99.5
HybridCBM	99.6	99.2

the concept head is under less classification pressure and can fit the concept labels more closely. Higher concept accuracy means closer agreement with the threshold labels; as the OOD results show, it does not by itself buy better detection.

Higher agreement with the labels does not by itself show that the bottleneck is free of *concept leakage*. We therefore train a linear classifier on the training split to predict the class from (i) the soft predicted concepts $\hat{\mathbf{c}} \in [0,1]^K$ and (ii) the same predictions thresholded at 0.5, and report the test-accuracy difference. This difference measures the additional predictive utility of the continuous values for these probes. It is small on CTU-IoT-23 for $\gamma \leq 0.5$ (five-seed means 0.001–0.013) but reaches $0.084 \pm .088$ at γ=1.0, where two of five seeds gain 0.17–0.21; on CIC-IoT-2023 it is 0.04–0.06 and grows rather than shrinks with γ (γ=0: $0.041 \pm .008$; γ=1.0: $0.058 \pm .008$).

A small difference does not establish semantic fidelity. Both probes use *predicted* concepts, and the thresholded predictions are not the declared predicates: where they disagree with the declared labels, the pattern of disagreement can itself carry class information. The predicate-replacement test of Section SXVI therefore uses the declared predicates directly.

SIII. Sensitivity to the Concept-Supervision Weight λ

Known-class F1 is invariant to the concept-supervision weight λ (± 0.002 on both datasets), so we fixed λ=0.5 in the main paper and varied γ. Raising λ raises concept accuracy on both datasets and moves OOD AUROC in dataset-dependent directions ($+0.09$ on CTU-IoT-23, -0.03 on CIC-IoT-2023 from λ=0.1 to 1.0), without affecting classification (Table S4).

TABLE S4: Sensitivity to the concept-supervision weight λ (JointCBM, γ=0, mean over five seeds; AUROC ± 1 std). These are the original-feature runs, reported as an ablation (Section SVII); they are not the known-class-only models of main Table III, and are shown only to establish the λ trend, which is qualitatively unaffected by the feature list.

	λ=0.1	λ=0.5	λ=1.0
CTU F1	0.9327	0.9325	0.9326
CTU concept-acc	0.650	0.856	0.905
CTU OOD AUROC	0.790±0.030	0.805±0.023	0.884±0.010
CIC F1	0.8201	0.8192	0.8189
CIC concept-acc	0.872	0.968	0.981
CIC OOD AUROC	0.752±0.015	0.748±0.012	0.724±0.021

SIV. Temporal (Timestamp-Ordered) Split on CTU-IoT-23

Random flow partitioning can leak device or session identity between train and test. A strict device- or session-disjoint split is harder for the CTU attacks, which come from very few source hosts (Okiru: 4, DDoS: 7, C&C-HeartBeat: 5 source IPs), so we instead build the known-class splits in time: we take the same per-class sample as the main split, order the flows of each known class by capture timestamp (timestamps are compared only within a class), and assign that class's earliest 70% to training, the next 15% to validation and the latest 15% to test (five seeds; within each class, test strictly follows train; the held-out unknown families are unchanged). Timestamp ordering does not by itself guarantee a session- or capture-disjoint split, individual sessions or captures may straddle the train/test boundary, so we claim temporal ordering, not capture-, session- or device-disjointness; the small number of source hosts makes a strict capture-disjoint split harder but we do not claim it impossible, and it remains untested. This protocol probes a property of the CTU-IoT-23 labelling: the *DDoS* family is temporally bimodal, a high-rate flood capture (September 2018, median Rate $\approx 9.3 \times 10^5$ pps) and near-zero-rate captures (December 2018 onward, median Rate <1), so a timestamp split can train on one DDoS regime and test on the other. This makes *every* model score F1 0.00 on DDoS on every seed; with the main split's 31-feature list on the same sample only two JointCBM γ=0.5 seeds keep DDoS (F1 0.65), so the collapse follows from the label's temporal structure rather than from feature selection (Section SVII).

Two effects separate. The *absolute* deterioration is large but comes from one class: four-class weighted F1 falls from 0.933 (random split) to 0.646–0.648 for every model but SequentialCBM ($F1_4$, main Table IV) because DDoS is lost; C&C-HeartBeat also falls, from 0.998 to about 0.79. This is a temporal-generalisation cost, not an interpretability one. The *between-model* gap is negligible: on the three coherent families every JointCBM variant, HybridCBM and NeSy-NIDS match the MLP in aggregate (Coh-3 0.862–0.863) while redistributing errors across classes (e.g. γ=0: Benign 0.841 vs. 0.958, Okiru 0.961 vs. 0.844). SequentialCBM is the exception, losing Benign ($F1_4$ 0.330, Coh-3 0.441). NeSy-NIDS gives the highest temporal-split AUROC (0.888 ± 0.025; 0.901 ± 0.030 with the

α-gate penalty of Section SI), and the JointCBM γ ordering shifts relative to the alternative split.

SV. Rule Selectivity and Threshold Drift

All values in this section are mean binarised activations (Eq. 5) per known class on the known-class validation split for the seed-0 model, the same data as Fig. 2. On CTU-IoT-23, several rules fire selectively but not deterministically (`cc_heartbeat_beaconing` at 0.78 and `incomplete_handshake` at 0.87 on C&C-HeartBeat, `asymmetric_flood` at 0.81 on DDoS and 0.13 on Benign, `ddos_high_rate` at 0.70 on DDoS and 0.21 on Okiru), leaving within-class variance for Mahalanobis to work with; others fire across several classes (`okiru_low_rate`: 0.79 on Okiru but 0.87 on Benign and 0.61 on C&C-HeartBeat). On CIC-IoT-2023 firing is near-deterministic (`syn_flood_ddos` at 1.00 on DDoS-SYN, `benign_low_rate` at 0.97 on Benign but also 0.91 on Mirai), within-class variance is close to zero, and Mahalanobis has little to separate on (AUROC 0.657). These two activation regimes coincide with the observed CTU-versus-CIC difference in Mahalanobis AUROC, but, as the main paper establishes, known-class activation geometry does not by itself govern detectability, so we report the pattern as descriptive, not causal (Fig. 2). The learned thresholds move away from their expert initial values (seed 0): on CTU-IoT-23 the `cc_heartbeat_beaconing` bounds move by up to -0.50, and on CIC-IoT-2023 by up to -0.88 on `syn_flood_ddos`; across the five seeds the per-condition drift vectors are correlated on CIC-IoT-2023 (mean pairwise r=0.73) but not on CTU-IoT-23 (r=0.05; script `nesy/run_fidelity.py`), so we read the CIC refinement as systematic and the CTU pattern as indicative only (main Fig. 3).

SVI. Configuration Sensitivity of the Geometry Correlations

Main Table XIII correlates each known-class statistic with OOD AUROC over 30 CBM models per dataset, which are five seeds of six configurations rather than 30 independent observations. Table S5 reports, on the same leak-free feature lists as the main table (CTU: known-class-only selection, Section SVII; CIC/TON: fixed lists), two further aggregations applied identically to all three corpora: the correlation over the six configuration means (r_6, n=6), and the range of the 30-model correlation when each configuration in turn is excluded (leave-one-configuration-out, LOCO). On CIC-IoT-2023 and TON_IoT the seven statistics agree in direction, and the six separability and variance statistics are strongly associated with AUROC at the configuration-mean level ($|r_6| \geq 0.93$ on both); closed-set F1 is weaker (r_6=0.63 and 0.73) and is the only statistic whose sign changes under LOCO there (on CIC). CTU-IoT-23 is different. All seven statistics take the opposite sign (r_6 from -0.82 to $+0.49$), and the LOCO ranges show why: every CTU range reaches its bound closest to zero when SequentialCBM is excluded, where the correlations fall to $|r| \leq 0.22$ and Fisher ratio, silhouette and closed-set F1 change sign. SequentialCBM combines the least separable concept space and the lowest closed-set F1 with one of the highest CTU AUROCs, so this single configuration sets the CTU direction. The unknown/known ratio, which uses the unknown flows, is positive everywhere (r_6 0.70–0.92). These are associations with Mahalanobis AUROC measured *within* each corpus, not a test that a known-class statistic predicts AUROC on an unseen corpus or selects between scorers; the strong r_6 values rest on six points, and the one selection procedure we tried, the leave-family-out proxy of Section SXI, did not reliably choose the scorer. The LOCO analysis thus locates the CTU reversal in one configuration rather than in the corpus.

TABLE S5: Configuration sensitivity of main Table XIII, on the same leak-free feature lists (CTU: known-class-only selection). r_6: Pearson r over the six configuration means. LOCO: range of the 30-model r when one configuration is excluded; on CTU the bound nearest zero is always the run without SequentialCBM. *Sign changes when SequentialCBM is excluded.

	CTU-IoT-23		CIC-IoT-2023		TON_IoT	
Statistic	r_6	LOCO	r_6	LOCO	r_6	LOCO
Within-class variance	$+0.49$	$[+0.01, +0.42]$	-0.96	$[-0.82, -0.66]$	-0.95	$[-0.54, -0.32]$
Fisher ratio	-0.45	$[-0.32, +0.02]^*$	$+0.96$	$[+0.56, +0.76]$	$+0.95$	$[+0.36, +0.54]$
Ridge-LDA	-0.50	$[-0.44, -0.08]$	$+0.95$	$[+0.31, +0.60]$	$+0.98$	$[+0.40, +0.57]$
Silhouette	-0.46	$[-0.42, +0.01]^*$	$+0.95$	$[+0.74, +0.85]$	$+0.94$	$[+0.40, +0.58]$
Nearest-centroid margin	-0.47	$[-0.43, -0.02]$	$+0.94$	$[+0.50, +0.74]$	$+0.94$	$[+0.32, +0.52]$
Concept entropy	-0.52	$[-0.47, -0.06]$	$+0.96$	$[+0.71, +0.83]$	$+0.93$	$[+0.25, +0.49]$
Closed-set F1	-0.82	$[-0.42, +0.22]^*$	$+0.63$	$[-0.28, +0.76]^*$	$+0.73$	$[+0.03, +0.42]$
Unknown/known ratio	$+0.78$	$[+0.59, +0.74]$	$+0.92$	$[+0.43, +0.64]$	$+0.70$	$[+0.70, +0.76]$

SVII. CTU-IoT-23 Feature Derivation

CTU-IoT-23 is distributed as packet captures with labelled Zeek `conn.log` files. For each capture we compute per-flow packet statistics with DPKT (field names follow the CIC-IoT-2023 feature set) and join each labelled Zeek record to the DPKT flow with the same source and destination IP and port whose start time lies in $[t_{\text{zeek}}, t_{\text{zeek}}+\text{duration}+1\,\text{s}]$; Zeek records without a match are dropped. Four ratios are derived from Zeek fields: byte ratio $\text{orig_bytes}/(\text{resp_bytes}+1)$, originator packet and byte rates $\text{orig_pkts}/(\text{duration}+1)$ and $\text{orig_ip_bytes}/(\text{duration}+1)$, and direction $\text{orig_bytes}-\text{resp_bytes}$. The model uses 34 numeric features (Zeek: duration, orig/resp bytes, packets and IP bytes; DPKT: Rate, Srate, IAT, Number, Tot_sum, Min, Max, AVG, Std, Variance, Magnitude, Radius, ack/syn/fin/rst counts, fin/rst flag numbers, and HTTP, HTTPS, SSH, IRC, Telnet indicators; plus the four derived ratios) and three Zeek categorical fields (conn_state, history, service) integer-encoded, 37 in total. Missing values are set to -1; features are not rescaled.

Feature-selection caveat. The 34 numeric features were chosen before this study as those covering 99% of cumulative random-forest importance, with a class-weighted forest trained on up to 10^6 flows of every labelled CTU family. That sample included the nine families later held out as unknown, so the CTU feature set is not blind to the held-out labels. The choice is fixed and shared by every model compared, but it may favour separability of the CTU unknowns in raw-feature space. CIC-IoT-2023 and TON_IoT use fixed feature lists (the 39 DPKT columns; 23 conn-log fields) without label-informed selection.

Known-class feature-selection control. Disclosing this is not enough, so we ran the control. We re-selected the CTU numeric features with the same random-forest procedure but on the known-class *training* rows only, with the nine held-out families excluded from selection. The known-only list keeps 28 numeric features (of 34): it drops the service and protocol indicators (HTTP, HTTPS, SSH, IRC), two counters (`resp_pkts`, `fin_count`) and several statistical or derived features (Radius, Std, Variance, `resp_ip_bytes`, `rst_flag_number`), and adds TCP, Protocol_Type, Time_To_Live and two TCP-flag indicators (`cwr`, `syn` flag numbers), 31 features in total with the three categorical fields. We drew a class-balanced split and ran every CTU comparison with this list (`ctuk`); as a paired baseline we ran the submitted 37-feature list on the *same* rows (`ctuko`), so `ctuko` vs `ctuk` isolates the selection source from the sample. The NeSy-NIDS rule features are expert-defined rather than RF-selected, so the two rule counters the known-only forest drops are restored and the rule indices remapped; only the neural branch and the CBM/MLP/DT/RF inputs change. The script `06_make_kfs_splits.py` and the drivers `cbm/run_kfs_control.py` and `nesy/run_kfs_control.sh` reproduce the table.

Table S6 reports the outcome over five seeds. The accuracy and scorer findings hold. The JointCBM variants, HybridCBM and NeSy-NIDS still match the MLP on known-class F1 (all 0.933); the two low-capacity models are unchanged (SequentialCBM 0.57, post-hoc CBM 0.58). Per-class Mahalanobis is the best CTU scorer for every architecture under both lists (Table S7), with the logit-energy score below chance for the MLP (0.39 vs. Mahalanobis 0.88). Concept-fidelity regularisation γ still has no monotone effect on OOD AUROC (JointCBM 0.82–0.83 and 0.81–0.84 across γ; Table S6). NeSy-NIDS, the strongest CTU detector, moves from 0.885 to 0.904.

Three OOD AUROC changes reach the 0.05 level read as material elsewhere, and all are gains under known-class-only selection. SequentialCBM rises from $0.782\pm.042$ to $0.872\pm.026$ ($\Delta = +0.090$, beyond its seed std), HybridCBM from $0.772\pm.073$ to $0.822\pm.055$ ($\Delta = +0.050$, below its own seed std) and Post-hoc CBM from 0.805 to 0.871; the MLP moves -0.020, the JointCBM variants by at most 0.03, and the DT and RF baselines gain 0.12 and 0.03. Removing the held-out labels from feature selection therefore does not flatter the CTU detection results; if anything the submitted list hurt the concept-space detectors.

The control also bears on the geometry reading. The CTU correlations of main Table XIII are opposite in sign to CIC-IoT-2023 and TON_IoT, and Section SVI traces the direction to SequentialCBM, the CBM configuration whose AUROC changes most with the feature list. Under the leak-free list no CTU association holds without that configuration, and we do not test whether any statistic predicts detectability on an unseen corpus or selects a scorer.

Per-split selection for the robustness experiments. The known-only selection above is fitted on the primary random split's known-class training rows, which is not automatically leak-free for the other CTU experiments. We therefore run the same random-forest selection separately on each: the alternative known set's training rows (excluding the now-held-out C&C-HeartBeat), the earliest 70% of the timestamp-ordered split, and each leave-one-family-out fold's reduced known set, so the family that experiment holds out, and any test-period flow, never informs its features. Each of these splits draws its own class-balanced sample; running the main split's 31-feature list on the same sample separates re-selection from the sample, and the accuracy, scorer and γ conclusions hold under both. One consequence is visible in the temporal split (Section SIV): the DDoS collapse occurs under both feature lists (bar two JointCBM seeds with the main-split list), so it is a property of the temporal split, not of feature selection.

TABLE S6: Known-class feature-selection control on CTU-IoT-23 (five seeds). `ctuko`: split with the submitted 37-feature list; `ctuk`: same split with features selected from known-class training rows only, so `ctuko` vs `ctuk` isolates the selection source from the sample. Known-class weighted F1 (mean) and per-class Mahalanobis OOD AUROC (mean±std). Accuracy is unchanged; known-class-only selection raises the OOD AUROC of SequentialCBM (+0.090, beyond its seed std), HybridCBM (+0.050) and Post-hoc CBM (+0.066).

	F1		OOD AUROC	
Model	`ctuko`	`ctuk`	`ctuko`	`ctuk`
MLP	0.933	0.933	0.894±.005	0.874±.002
JointCBM (γ=0)	0.933	0.933	0.817±.031	0.823±.044
JointCBM (γ=0.1)	0.933	0.933	0.821±.033	0.805±.035
JointCBM (γ=0.5)	0.933	0.933	0.821±.030	0.844±.036
JointCBM (γ=1)	0.933	0.933	0.833±.029	0.834±.050
SequentialCBM	0.570	0.567	0.782±.042	0.872±.026
HybridCBM	0.933	0.933	0.772±.073	0.822±.055
NeSy-NIDS	0.933	0.933	0.885±.020	0.904±.014
Decision tree	0.934	0.934	0.557±.000	0.679±.000
Random forest	0.934	0.934	0.630±.002	0.657±.034
Post-hoc CBM	0.576	0.576	0.805±.030	0.871±.024

TABLE S7: Scorer comparison under the known-only feature selection (`ctuk`): OOD AUROC (mean±std, five seeds) for per-class Mahalanobis on the concept space, the logit-space energy score, and their z-normalised average (combined). Per-class Mahalanobis is best for every architecture, and the logit-energy score reaches at most 0.61 (main Table VII gives the CIC comparison, where the ranking depends on architecture).

Architecture	Mahalanobis	Energy	Combined
MLP	**0.879±.001**	0.386±.073	0.703±.061
JointCBM (γ=0)	**0.830±.044**	0.476±.111	0.632±.068
JointCBM (γ=0.1)	**0.812±.034**	0.494±.136	0.625±.097
JointCBM (γ=0.5)	**0.851±.037**	0.570±.138	0.718±.105
JointCBM (γ=1)	**0.842±.049**	0.612±.124	0.708±.100
SequentialCBM	**0.875±.026**	0.604±.085	0.757±.018
HybridCBM	**0.827±.055**	0.365±.047	0.613±.047

SVIII. NeSy-NIDS Rule Bank and Initial Thresholds

Table S8 gives the complete eight-rule bank for each dataset used by NeSy-NIDS (Section III-C of the main paper lists only representative rules). Each rule is a conjunction of threshold conditions on named flow features; the thresholds shown are the expert *initial* values $\boldsymbol{\theta}^{(0)}$, which are then refined by gradient

TABLE S8: Complete NeSy-NIDS rule bank (R=8 per dataset): target class and the conjunction of conditions at the expert *initial* thresholds (refined during training). CTU-IoT-23 in raw units; CIC-IoT-2023 in z-scored units.

Rule	Target	Condition (initial threshold)
CTU-IoT-23 (raw units)		
ddos_high_rate	DDoS	Rate>1000
cc_heartbeat_beaconing	C&C-HB	1.5<IAT<5.0 ∧ 0.1<Rate<5.0
benign_single_probe	Benign	orig_pkts<2∧ Rate<100
okiru_low_rate	Okiru	Rate<500∧ duration<2.5
asymmetric_flood	DDoS	resp_pkts<1∧ orig_pkts>1
persistent_connection	C&C-HB	duration>2.5
incomplete_handshake	C&C-HB	syn_count>2∧ fin_count<1
ddos_short_burst	DDoS	Rate>1000∧ duration<0.1
CIC-IoT-2023 (z-scored units)		
syn_flood_ddos	DDoS-SYN	rate>0.15∧ syn_count>1.5
udp_mirai_flood	Mirai	tcp<−1.0∧ rate>0.0
recon_port_scan	Recon	rst_count>1.5
vuln_scan	VulnScan	fin_count>1.0∧ rate>0.0
benign_low_rate	Benign	rate<0.0∧ syn_count<0.0
large_payload_flood	Benign	avg>0.4∧ udp>0.0
high_entropy_scan	Recon	variance>0.5∧ rst_count>0.5
short_connection_flood	DDoS-SYN	iat<−0.003∧ rate>0.1

descent through the k-annealed gates. The full feature lists (per-dataset `feature_cols`), the deterministic concept/rule threshold values, and the train/val/test split indices are released with the code (see the code-availability statement), so every representation in the paper is reproducible.

SIX. Threshold Gradient

Training uses the soft activations of main Eq. 4 throughout, so every threshold receives a gradient. For a greater-than condition $i \in \mathcal{I}_r^{>}$ of rule r,

$$\frac{\partial \mathcal{L}}{\partial \theta_i} = \frac{\partial \mathcal{L}}{\partial a_r} \cdot \Big(-k(t)\, g_i^{>} (1 - g_i^{>}) \prod_{j \neq i} g_j \Big), \qquad (1)$$

where $g_i^{>} = \sigma(k(t)(x_i - \theta_i))$ and g_j are the other gates of the rule; the sign flips for less-than conditions. As $k(t)$ grows, $g_i^{>}(1-g_i^{>})$ vanishes for gates far from their threshold, so late in training only borderline conditions keep moving. The straight-through estimator is not used in training.

SX. OOD AUROC versus Closed-Set F1

Figure S2 plots every model of main Table III in the F1–AUROC plane; main Section IV-D discusses what it shows for the closed-set/open-set relationship.

SXI. Leave-Family-Out Scorer Selection

The measured associations between known-class statistics and OOD AUROC vary across datasets (main Table XIII); whether a scorer can be *selected* in advance is a separate question, which we test directly with a *behavioural* proxy: hold out one known attack family, retrain JointCBM (γ=0) on the reduced known set, and keep the scorer (per-class Mahalanobis vs logit-space energy) that best detects the held-out family, then ask whether that choice matches the scorer actually best on the true unknowns (Mahalanobis on CTU-IoT-23, energy

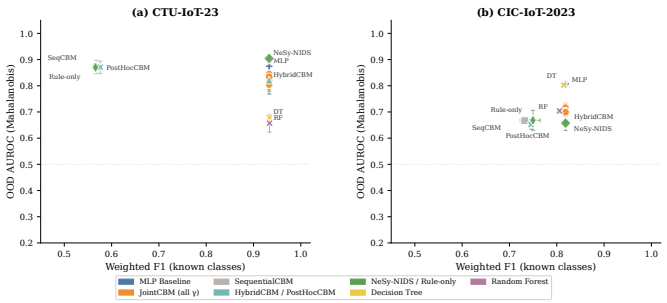


Fig. S2: OOD AUROC vs. known-class weighted F1 for all models on CTU-IoT-23 (left) and CIC-IoT-2023 (right), using the values of main Table III. Error bars are ±1 std over five seeds; every model uses five seeds (DT's raw-feature score has no seed variance).

on CIC-IoT-2023). We repeat for every known attack family × five seeds (main Table VIII).

The proxy is only *conditionally* reliable. On CTU-IoT-23 the family-majority vote recovers Mahalanobis (two of three families, $9/15$ folds); the C&C-HeartBeat and DDoS folds favour it, and only Okiru misfires (Mahalanobis cannot detect held-out Okiru, 0.22). On CIC-IoT-2023 the family-majority vote selects Mahalanobis for all four families ($15/20$ individual folds, $5/20$ energy) and so never recovers energy: held-out known families remain separable in activation space (Mahalanobis 0.70–0.86), so the proxy selects Mahalanobis, yet the true CIC unknowns overlap the known boundary and need energy. Held-out-known thus over-approximates activation-space detectability exactly on the corpus where the scorer must switch. In these experiments, therefore, this leave-family-out proxy did not consistently identify the best scorer; selecting it reliably would require observing the novel regime, which we do not demonstrate a priori. This is a further instance of the paper's negative result, here for scorer *selection* rather than scorer performance. Driver: `cbm/run_lofo.py`. A second rule that needs no retraining, based on synthetic outliers, is evaluated in Section SXII.

SXII. Transfer and Selection Tests

Do known-class statistics transfer across corpora? Main Table XIII reports within-corpus associations. To test prediction on an unseen corpus, we fit AUROC $= a + b\,s$ by least squares on the 60 CBM concept-space models of two corpora, for each known-class statistic s, and apply it to the 30 models of the held-out third (Table S9). The prediction beats the training-corpora mean AUROC in 3 of 21 cases, ranks the held-out models in the right direction (positive Spearman correlation) in 7 of 21, and identifies the best configuration in 1 of 21. The fitted slope is dominated by the differences between corpora (the dataset means are non-monotone, main Section IV-D), so it often has the wrong sign even for CIC-IoT-2023 and TON_IoT, where the within-corpus direction agrees; statistics on different scales, such as ridge-LDA separability, also transfer poorly in absolute terms. Driver: `cbm/run_geometry_transfer.py`.

Scorer selection from synthetic outliers. The leave-family-out proxy (Section SXI) needs retraining. As a second rule that

TABLE S9: Leave-one-corpus-out prediction of OOD AUROC from each known-class statistic (fitted on the other two corpora, 60 models; evaluated on the 30 held-out models). Slope: fitted b; MAE: mean absolute AUROC error on the held-out corpus; ρ: Spearman correlation of predicted and true AUROC; Best: the predicted best configuration is the true best.

Statistic	Slope	MAE	ρ	Best
Held out: CTU (training-mean MAE 0.167)				
Within-class variance	−1.774	0.198	−0.36	no
Fisher ratio	+0.065	0.169	−0.34	no
Ridge-LDA	+0.000	0.180	−0.38	no
Silhouette	+0.336	0.149	−0.34	no
Nearest-centroid margin	+0.201	0.152	−0.28	no
Concept entropy	+0.006	0.167	−0.38	no
Closed-set F1	−0.022	0.168	+0.17	yes
Held out: CIC (training-mean MAE 0.045)				
Within-class variance	+0.616	0.030	−0.66	no
Fisher ratio	+0.060	0.075	+0.60	no
Ridge-LDA	+0.003	0.671	+0.41	no
Silhouette	+0.263	0.069	+0.75	no
Nearest-centroid margin	+0.247	0.083	+0.54	no
Concept entropy	−0.415	0.064	−0.70	no
Closed-set F1	−0.006	0.045	−0.17	no
Held out: TON (training-mean MAE 0.123)				
Within-class variance	+2.001	0.146	−0.46	no
Fisher ratio	−0.045	0.145	−0.51	no
Ridge-LDA	−0.000	0.153	−0.58	no
Silhouette	−0.092	0.136	−0.43	no
Nearest-centroid margin	−0.113	0.149	−0.39	no
Concept entropy	+0.113	0.130	+0.37	no
Closed-set F1	+0.128	0.125	+0.30	no

TABLE S10: Scorer selection from synthetic outliers (five seeds). True: OOD AUROC on the held-out unknown families; Perm.: AUROC separating validation flows from feature-permuted copies; Correct: seeds on which the rule picks the scorer that is better on the true unknowns (permutation / midpoint rule).

	True		Perm.		Correct	
Architecture	Mahal.	Energy	Mahal.	Energy	Perm.	Mid.
CTU-IoT-23						
MLP	0.879	0.386	0.984	0.766	5/5	4/5
JointCBM (γ=0)	0.830	0.476	0.802	0.703	4/5	4/5
JointCBM (γ=0.1)	0.812	0.494	0.795	0.747	3/5	4/5
JointCBM (γ=0.5)	0.851	0.570	0.823	0.747	4/5	5/5
JointCBM (γ=1.0)	0.842	0.612	0.853	0.786	5/5	5/5
SequentialCBM	0.875	0.604	0.644	0.585	4/5	4/5
HybridCBM	0.827	0.365	0.651	0.801	0/5	1/5
CIC-IoT-2023						
MLP	0.807	0.615	0.974	0.655	5/5	5/5
JointCBM (γ=0)	0.722	0.734	0.769	0.803	3/5	3/5
JointCBM (γ=0.1)	0.705	0.732	0.779	0.821	4/5	2/5
JointCBM (γ=0.5)	0.706	0.757	0.771	0.826	5/5	1/5
JointCBM (γ=1.0)	0.702	0.769	0.778	0.820	5/5	0/5
SequentialCBM	0.665	0.636	0.723	0.689	4/5	5/5
HybridCBM	0.670	0.629	0.678	0.732	0/5	5/5
TON_IoT						
MLP	0.550	0.662	0.859	0.523	0/5	0/5
JointCBM (γ=0)	0.644	0.713	0.740	0.621	1/5	0/5
JointCBM (γ=0.1)	0.640	0.674	0.758	0.624	3/5	2/5
JointCBM (γ=0.5)	0.641	0.694	0.770	0.630	2/5	2/5
JointCBM (γ=1.0)	0.642	0.706	0.758	0.637	1/5	1/5
SequentialCBM	0.603	0.640	0.730	0.637	0/5	2/5
HybridCBM	0.615	0.641	0.747	0.548	2/5	2/5

uses only known traffic, we build synthetic outliers from known validation flows and select the scorer, per-class Mahalanobis or logit energy, that better separates them from a disjoint set of validation flows. The primary construction, fixed before evaluation, permutes each feature independently across flows, keeping every marginal and breaking the joint structure of real traffic; a secondary construction uses midpoints of random pairs of validation flows. The choice is compared, per model and seed, with the scorer that is better on the true unknowns, on the same evaluation subsets as main Table VII (Table S10). The permutation rule is right for $25/35$ CTU, $26/35$ CIC and $9/35$ TON_IoT models ($60/105$); the midpoint rule for $27/35$, $21/35$ and $9/35$ ($57/105$). Always choosing Mahalanobis is right for $62/105$ and always choosing energy for $43/105$, and the wrong choices of the permutation rule cost 0.066 AUROC on average. Both rules prefer Mahalanobis on TON_IoT, where energy is better for every architecture, so neither selects the scorer reliably. Driver: `cbm/run_scorer_selection.py`.

SXIII. Open-Set (C+1) and Per-Family Evaluation

Protocol. We evaluate the deployed $(C+1)$-class detector at a reject threshold τ calibrated to 5% FPR on the known validation split. The scorer is per-class Mahalanobis on the model's scoring space: the concept activations for CBM variants, the binary rule activations for NeSy-NIDS, and the penultimate embedding for the MLP. Ground truth has C known classes plus a single "unknown" class into which *all* held-out families are merged; a flow is predicted "unknown" if its score exceeds τ, else the $\arg\max$ known-class logit. Evaluation uses $5{,}000$ known test flows (all seeds) and, per unknown family, up to $3{,}000$ flows: 9 families merged into "unknown" on CTU-IoT-23 and 29 on CIC-IoT-2023. Main Table IX reports, over five seeds, the known-class rejection rate (target $\approx 5\%$), the open-set macro-F1 over the $C+1$ classes, and the equal-family-weighted mean AUROC.

The known rejection rate is close to target (4.2–5.8%). Across seeds the coefficient of variation of τ is at most 0.11 for every model (0.09 and 0.11 for NeSy-NIDS on CTU-IoT-23 and CIC-IoT-2023). Open-set macro-F1 is modest on CTU-IoT-23 (0.49–0.79) and low on CIC-IoT-2023 (0.13–0.27). The mixture is essential to reading this number. The evaluation pool is heavily unknown-skewed, for NeSy-NIDS on CIC-IoT-2023 (Table S11, seed 0) it is $5{,}000$ known against $84{,}514$ unknown flows, about 16.9 unknowns per known flow, and only $15{,}646$ (18.5%) of the pooled unknown flows exceed τ. Macro-F1 weights every class equally, so it is *not* dominated by the unknown class; in that matrix the unknown class in fact has the *highest* per-class F1 (≈ 0.31). What pulls macro-F1 down is a prevalence effect on the *known* classes: the $\sim 81\%$ of unknown flows that are not rejected are absorbed into known classes (most often VulnerabilityScan, Mirai, and DDoS-SYN), and because unknowns outnumber knowns $\sim$17:1 these false positives collapse known-class *precision* (Benign precision ≈ 0.07, so F1 ≈ 0.13) even though known *recall* stays high for accepted flows. The low unknown recall is a genuine weakness, but the small macro-F1 is as much a prevalence artefact as a detection failure: reweighting the same confusion matrix to

TABLE S11: $(C+1)$ confusion matrix, NeSy-NIDS on CIC-IoT-2023 (seed 0; row = true, column = predicted). All 29 held-out families are merged into the single "unknown/reject" row. VulnSc.: VulnerabilityScan.

true \ pred	Benign	DDoS-SYN	Mirai	Recon	VulnSc.	reject
Benign	765	0	0	70	161	1
DDoS-SYN	0	911	0	0	0	37
Mirai	0	0	1073	0	0	3
Recon-PortScan	155	0	0	349	265	175
VulnScan	126	0	0	95	801	13
unknown	9633	13459	16918	6349	22509	15646

equal known/unknown totals, with all conditional prediction rates unchanged, raises macro-F1 from ≈ 0.14 to ≈ 0.51, an arithmetic sensitivity check, not a better model. We therefore treat the operating-point macro-F1 as prevalence-sensitive and base the detection claim on the (prevalence-free) per-family AUROC of Table S12.

Table S12 shows per-family AUROC and unknown recall at τ for JointCBM (γ=0) and NeSy-NIDS; averaging its recall column over families with equal weight gives the family-averaged recall of main Table IX (0.51/0.36 for JointCBM, 0.83/0.20 for NeSy-NIDS on CTU/CIC). Pooled AUROC masks extreme per-family variation: for JointCBM detection is near-perfect for several volumetric DDoS families (AUROC > 0.97, recall > 0.8) but below chance for others (CIC Mirai-greip_flood 0.15; DDoS-UDP_Fragmentation 0.24; DDoS-SynonymousIP 0.48) and weak for most injection/scan/web families (0.58–0.71, recall $\leq$0.22), and NeSy-NIDS catches a different subset (Table S12). The equal-family-weighted mean AUROC (0.818 CTU, 0.722 CIC) is close to the pooled AUROC of main Table III (0.823 CTU, 0.717 CIC), but per-family detection ranges from near-perfect to near-chance for both models, so the pooled figure should not be read as a uniform per-family detection rate.

TABLE S12: Per-family OOD AUROC and unknown recall at τ (5% FPR) for JointCBM (γ=0) and NeSy-NIDS, mean over five seeds. Detection varies enormously by family for both models, and the two disagree on which families they catch.

	JointCBM		NeSy-NIDS	
Held-out family	AUROC	Rec.	AUROC	Rec.
CTU-IoT-23				
C&C	0.832	0.33	0.864	0.29
Attack	0.812	0.45	0.995	0.99
C&C-HeartBeat-Attack	0.819	0.55	0.993	0.94
C&C-FileDownload	0.712	0.42	0.996	1.00
C&C-Torii	0.812	0.27	0.844	0.28
C&C-Mirai	0.994	1.00	0.994	1.00
C&C-HeartBeat-FileDownload	0.701	0.40	0.995	1.00
FileDownload	0.715	0.40	0.995	1.00
Okiru-Attack	0.964	0.80	0.979	0.93
CIC-IoT-2023				
DDoS-UDP_Flood	0.978	0.847	0.888	0.453
DDoS-TCP_Flood	0.902	0.390	0.830	0.107
DDoS-ICMP_Flood	0.949	0.638	0.900	0.481
DDoS-RSTFINFlood	0.974	0.816	0.993	0.958
DDoS-PSHACK_Flood	0.993	0.993	0.848	0.178
DDoS-SynonymousIP_Flood	0.475	0.002	0.350	0.015
DDoS-ACK_Fragmentation	0.962	0.821	0.516	0.055
DDoS-UDP_Fragmentation	0.237	0.024	0.892	0.343
DDoS-HTTP_Flood	0.969	0.881	0.868	0.606
DDoS-SlowLoris	0.977	0.898	0.705	0.284
DDoS-ICMP_Fragmentation	0.636	0.219	0.459	0.007
DoS-SYN_Flood	0.543	0.088	0.452	0.124
DoS-UDP_Flood	0.970	0.761	0.839	0.365
DoS-TCP_Flood	0.883	0.333	0.809	0.077
DoS-HTTP_Flood	0.975	0.907	0.897	0.697
Mirai-greip_flood	0.148	0.021	0.551	0.005
Mirai-udpplain	0.632	0.029	0.937	0.628
Recon-HostDiscovery	0.711	0.223	0.560	0.098
Recon-OSScan	0.661	0.066	0.537	0.074
Recon-PingSweep	0.635	0.091	0.454	0.014
MITM-ArpSpoofing	0.742	0.245	0.585	0.072
DNS_Spoofing	0.721	0.291	0.547	0.060
DictionaryBruteForce	0.582	0.065	0.441	0.007
BrowserHijacking	0.663	0.117	0.496	0.026
CommandInjection	0.647	0.156	0.487	0.008
XSS	0.597	0.103	0.441	0.006
SqlInjection	0.577	0.077	0.421	0.003
Backdoor_Malware	0.607	0.120	0.446	0.006
Uploading_Attack	0.596	0.128	0.431	0.004

SXIV. Per-Class F1 on the Random Split

A single weighted F1 can hide minority-class failure. Table S13 gives per-class F1 on the known test split (random split, mean over five seeds) for the MLP and JointCBM (γ=0). The two agree within 0.002 for every class, so under matched-distribution evaluation this model shows no measurable per-class difference from the MLP; the other variants are not tabulated here. The weighted F1 nonetheless masks real spread: CTU Okiru is 0.86 and DDoS 0.90 (vs 0.98–1.00 for Benign and C&C-HeartBeat) and CIC Recon-PortScan is only 0.64, VulnerabilityScan 0.70 and Benign 0.76.

SXV. Joint (λ, γ) Sweep

λ (BCE concept supervision) and γ (added L2 concept fidelity) both push the predicted concepts toward the same labels, so we sweep them jointly (JointCBM, three seeds per cell) rather than as two 1-D slices, to check that the conclusions are not an artifact of holding one fixed (Table S14; CTU on the original-feature split as in Table S4, CIC on the main split). Three points hold in 2-D. (i) γ is not a reliable lever: its effect on OOD AUROC is small ($|\Delta| \leq 0.046$) and its *sign depends on* λ, positive at λ=0.5 on CTU-IoT-23 (+0.046) but negative at λ=1.0 (-0.027), and negative throughout CIC-IoT-2023. (ii) λ has a larger effect, but only on CTU-IoT-23 (down the γ=0 column, +0.092), not on CIC-IoT-2023 (-0.033); this is why the 1-D λ sweep looked larger than the γ sweep. (iii) Known-class F1 is nearly constant across the grid (0.816–0.821 on CIC; 0.932–0.933 on CTU) with a single exception, CTU λ=1.0, γ=0.5, seed 0, where F1 fell to 0.748 (the other two seeds of that cell are 0.933/0.933); this run is *included* in the reported cell mean. It shows that heavy dual regularisation can destabilise training; in the other 53 runs of the grid known-class F1 stays within the ranges above. The λ–γ interaction (a sign flip in γ's effect across λ) is one more instance of no single tunable factor governing open-set detectability.

TABLE S13: Per-class known-test F1 on the random split (mean over five seeds). MLP vs JointCBM (γ=0); values agree to within 0.002.

Dataset	Class	F1 (MLP / JointCBM)
CTU	Benign	0.977 / 0.976
CTU	C&C-HeartBeat	0.999 / 0.998
CTU	DDoS	0.900 / 0.900
CTU	Okiru	0.857 / 0.857
CIC	Benign	0.760 / 0.759
CIC	DDoS-SYN_Flood	0.999 / 0.999
CIC	Mirai-greeth_flood	0.999 / 0.999
CIC	Recon-PortScan	0.636 / 0.638
CIC	VulnerabilityScan	0.699 / 0.697

TABLE S14: Joint (λ, γ) sweep: JointCBM OOD AUROC (mean over three seeds; std in parentheses). λ down rows, γ across columns.

	γ=0	γ=0.5	γ=1.0
CTU-IoT-23			
λ=0.1	0.798 (.037)	0.792 (.043)	0.817 (.022)
λ=0.5	0.817 (.017)	0.868 (.027)	0.863 (.009)
λ=1.0	0.890 (.007)	0.876 (.007)	0.863 (.026)
CIC-IoT-2023			
λ=0.1	0.747 (.020)	0.727 (.026)	0.710 (.024)
λ=0.5	0.729 (.014)	0.707 (.006)	0.702 (.018)
λ=1.0	0.714 (.016)	0.715 (.006)	0.706 (.005)

SXVI. Concept-Fidelity Intervention (CBM)

Comparing soft concepts with their binarised values cannot establish semantic fidelity, since both can contain concept errors. Because our concepts are deterministic functions of the input, we test directly how much the JointCBM's accuracy depends on *deviations* from the declared predicates. We (i) run the model normally ($F1_{pred}$), (ii) intervene by replacing the predicted concepts with the *true* binary predicates $\mathbf{c}_{gt}$ and passing them through the model's own learned classifier head ($F1_{gt}$), and (iii) fit a fresh logistic regression on $\mathbf{c}_{gt}$ alone ($F1_{LR}$, how well this linear classifier can classify from the declared predicates; it is not an upper bound on the information they contain).

The gap is large (Table S15). On CTU-IoT-23 the model scores F1 0.93 but its classifier on the true predicates scores only 0.46–0.49, and a logistic regression on the declared predicates alone reaches 0.58; on CIC-IoT-2023 the model scores 0.82 versus 0.62–0.65 from true predicates and 0.72 from a logistic regression on the predicates alone. So the eight declared predicates do not, under the evaluated linear classifiers, account for the CBM's known-class accuracy. Because the intervention alters the classifier's input distribution, this gap tests decision fidelity under predicate replacement; it does not attribute a fixed fraction of accuracy to a source. What it shows is that the learned concept values, whether through their continuous magnitudes or through their disagreements with the declared labels, carry decision-relevant information that the declared predicates do not. On CTU-IoT-23 the soft-versus-thresholded probe difference is small for $\gamma \leq 0.5$ (Section SII), so there most of that information survives thresholding. The mean gap shrinks slightly with γ (CTU $0.476 \rightarrow 0.441$), but by about one seed std (0.02–0.04).

TABLE S15: Concept-fidelity intervention (JointCBM, five seeds). $F1_{pred}$: normal; $F1_{gt}$: true predicates through the learned classifier; $F1_{LR}$: logistic regression on true predicates alone (deterministic, so no seed std). Values are mean (std) over five seeds. The intervention changes the classifier's input distribution, so the gap tests decision fidelity under predicate replacement, not a decomposition of accuracy.

Data	γ	$F1_{pred}$	$F1_{gt}$	gap
CTU	0	0.933 (.000)	0.457 (.024)	0.476 (.024)
CTU	0.1	0.933 (.000)	0.489 (.029)	0.444 (.029)
CTU	0.5	0.933 (.000)	0.483 (.016)	0.450 (.015)
CTU	1.0	0.933 (.000)	0.492 (.038)	0.441 (.038)
declared predicates alone (logistic regression): F1 = 0.575				
CIC	0	0.818 (.001)	0.624 (.021)	0.194 (.021)
CIC	0.1	0.819 (.001)	0.642 (.040)	0.177 (.040)
CIC	0.5	0.818 (.001)	0.645 (.028)	0.173 (.028)
CIC	1.0	0.818 (.001)	0.639 (.035)	0.179 (.036)
declared predicates alone (logistic regression): F1 = 0.718				